Routing Visual Evidence: Three-Way Factorization for Cross-Domain Audio-Visual Deepfake Detection

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Abstract

Audio-visual deepfake detectors trained on large source datasets often collapse to domain-specific shortcuts rather than transferable manipulation evidence. We propose TRIROUTE, a three-way factorized audio-visual detector that separates shared cross-modal content, modality-unique manipulation cues, and residual domain-sensitive signal. The prediction head reads only the task-relevant shared and unique factors, while domain push-pull discourages easy domain prediction from this branch and concentrates nuisance information in the residual branch. On a matched AV-Deepfake1M→FakeAVCeleb benchmark, TRIROUTE improves AUC on the most informative category, FakeVideo-RealAudio (FV-RA), from 0.500 ± 0.047 for a two-way baseline to 0.659 ± 0.079. Crucially, adding domain adversarial pressure to a two-way representation without the three-way factorization decreases FV-RA AUC to 0.450, demonstrating that the factorized structure is the essential enabling condition. Subspace probes identify the visual-unique channel as the strongest OOD carrier, and head ablations show that masking this channel drops FV-RA from 0.721 to 0.366. These results support a routing-based rather than invariance-based explanation of the cross-domain gain.

1 Introduction

Audio-visual deepfakes have become harder to detect because modern generators can jointly manipulate lip motion, facial appearance, and speech with high realism. Existing detectors are often trained as standard supervised binary classifiers. While such models can report high scores on the dataset they are trained on, they frequently overfit to generator-specific fingerprints, compression artifacts, or annotation shortcuts instead of learning transferable manipulation evidence. This problem is especially acute in cross-domain evaluation, where the train and test distributions differ in source identities, synthesis pipelines, codecs, frame rates, and curation procedures.

Recent work has shown that audio-visual synchronization and speech-aware representations can improve robustness by anchoring detection to cross-modal consistency rather than static image artifacts [4, 8, 11]. However, even synchronization-based approaches can still entangle three qualitatively different signals: *what* is being spoken, *whether* the visible motion is manipulated, and *which domain* the sample comes from. In practice, these factors interact. A detector that does not explicitly separate them may use domain-specific shortcuts whenever such cues correlate with labels.

We address this issue with TRIROUTE, a model that factorizes audio-visual evidence into three roles: shared cross-modal content, modality-unique manipulation cues, and residual domain-sensitive signal. The shared factor is trained to remain aligned with speech semantics. The residual factor is trained to absorb dataset-specific nuisance information. The prediction head reads only the task-relevant shared and unique factors, so the model is encouraged to solve detection without relying on the residual shortcut branch. This design separates “useful for cross-domain detection” from “useful only for identifying the dataset.”

Contributions. We make three contributions.

- We propose TRIROUTE, a three-way factorized audio-visual detector that separates shared content, modality-unique manipulation evidence, and residual domain-sensitive signal, and applies domain push-pull to change which evidence the detector uses under shift.
- We introduce a joint objective combining clip-level detection, audio-visual alignment, orthogonality between task and residual factors, and adversarial domain suppression of the task branch.
- We provide direct evidence that the improvement in cross-domain visual-fake detection requires the factorized architecture itself: domain adversarial training without three-way factorization degrades FV-RA AUC (from 0.500 to 0.450), while TRIROUTE improves it to 0.659. Subspace probes and head ablations quantify the routing mechanism that drives this gain.

2 Related Work

Audio-visual deepfake detection. Multimodal deepfake detection leverages the fact that realistic speech, lip motion, and facial dynamics must stay mutually consistent over time. Early work explored direct audio-visual fusion and synchronization cues [12, 2]. More recent methods use self-supervised anomaly detection or speech representation learning to detect inconsistencies without relying heavily on fake samples during training [4, 8, 11]. These approaches are more robust than purely artifact-based detectors, but their representations may still mix semantic content with domain bias.

Evaluation protocols. Most audio-visual deepfake work still reports in-dataset or within-benchmark splits. FakeAVCeleb-based evaluation commonly uses train/test splits and leave-one-manipulation protocols [4, 9], while AV-Deepfake1M and DiMoDif report official or in-dataset benchmarks on AV1M, LAV-DF, and FAVC [1, 7]. The closest prior cross-dataset bridges are AVH-Align, which reports supervised AV1M→FAVC scores while diagnosing leading-silence shortcuts, and DiMoDif, which includes an aggregate cross-dataset matrix over FAVC, LAV-DF, and AV1M [11, 7]. Our contribution is therefore not the existence of FakeAVCeleb as a target dataset, but a matched AV1M→FAVC stress test with modality-category reporting and routing analyses focused on FV-RA.

Speech-aware multimodal representations. Large-scale audio-visual speech models, such as AV-HuBERT [10], learn strong multimodal features that capture both local synchronization and longer-term speech structure. Such representations have proven useful beyond recognition, including forensic analysis and synchronization-based detection [8, 11]. Our method uses the same intuition, but adds an explicit factorization layer so that the downstream detector can discard nuisance factors instead of letting them leak into the prediction head.

Domain generalization and factorized representations. Domain-adversarial learning [5] and factorized representations are established tools for improving transfer beyond the training distribution. In deepfake detection, dataset-specific artifacts often dominate the supervision signal, so models benefit from architectures that separate task-relevant evidence from nuisance. Our formulation adapts this principle to multimodal forensics by assigning domain information to a dedicated residual branch and discouraging the prediction head from depending on it.

3 Problem Setting

We study cross-dataset audio-visual deepfake detection. Let $\mathcal{D} = \{(x_i^v, x_i^a, y_i, d_i)\}_{i=1}^N$ denote training samples where x_i^v is a sequence of visual observations, x_i^a is the synchronized audio, $y_i \in \{0, 1\}$ is the video-level real/fake label, and $d_i \in \{1, \dots, M\}$ indicates the source domain or dataset. Training is dominated by source-domain clips from AV-Deepfake1M. In the full domain-aware setting, we may additionally use real-only target-domain clips for auxiliary domain supervision, while target-domain fake labels and all test clips remain untouched.

Our goal is to learn a scoring function $f(x^v, x^a)$ that remains predictive when the nuisance statistics associated with d shift at test time. We therefore seek a representation in which manipulation evidence remains discriminative for y, while predictable but non-transferable signals are isolated rather than implicitly used by the classifier. We do not claim that the learned task branch becomes fully domain-invariant; the claim we test is weaker and more mechanism-focused: factorization changes representation usage and downstream routing under shift.

4 Method

4.1 Overview

Figure 1 gives a conceptual overview of TriRoute. Starting from frozen AV-HuBERT features $z_v, z_a \in \mathbb{R}^{1024}$, we first extract shared cross-modal factors and modality-specific hidden states:

$$\big[s_v, h_v\big] = F_v(z_v), \qquad \big[s_a, h_a\big] = F_a(z_a), \tag{1}$$

where s_v and s_a denote shared task-relevant factors and h_v, h_a collect the remaining modality-specific signal. We then split the modality-specific states into unique task-relevant and residual components:

$$\big[u_v, r_v\big] = G_v(h_v), \qquad \big[u_a, r_a\big] = G_a(h_a). \tag{2}$$

Here u_v and u_a are modality-unique task factors, while r_v and r_a form the residual branch that is allowed to absorb shortcut and domain signal. The task representation and residual representation are

$$Z_{\text{task}} = \text{cat}(s_v, u_v, s_a, u_a), \qquad Z_{\text{res}} = \text{cat}(r_v, r_a). \tag{3}$$

For readability, we refer to the shared factor as *content*, the visual-unique task factor as the key *manipulation* carrier for FV-RA, and the residual branch as the *domain* factor.

4.2 Three-Way Factorization

Manipulation factor. The manipulation factor is intended to encode modality-unique evidence that cannot be explained by authentic cross-modal correspondence. In practice, the visual-unique component u_v is the most important carrier for the visual-fake category FV-RA, while u_a supports audio-fake cases. These unique components feed the task head together with the shared factors.

Content factor. The content factor captures shared speech semantics and identity-preserving motion that should remain useful across domains. We align the shared factors across modalities using an audio-visual alignment objective:

$$\mathcal{L}_{\text{align}} = -\sum_{t=1}^{T} \log \frac{\exp(\text{sim}(s_t^v, s_t^a)/\tau)}{\sum_{j \in \mathcal{N}(t)} \exp(\text{sim}(s_t^v, s_j^a)/\tau)}, \tag{4}$$

where $\text{sim}(\cdot,\cdot)$ is cosine similarity, τ is a temperature, and $\mathcal{N}(t)$ is a local temporal window.

Domain factor. The residual branch is explicitly optimized to predict the source domain:

$$\mathcal{L}_{\text{dom}} = \text{CE}(C_d(Z_{\text{res}}), d). \tag{5}$$

This branch acts as an information sink for codec, resolution, acquisition, and dataset-construction artifacts.

4.3 Detection and Domain Push-Pull Objectives

We predict a clip-level fake probability from the task representation:

$$\hat{y} = C_y(Z_{\text{task}}), \tag{6}$$

and optimize binary cross-entropy:

$$\mathcal{L}_{\text{fake}} = \text{BCE}(\hat{y}, y). \tag{7}$$

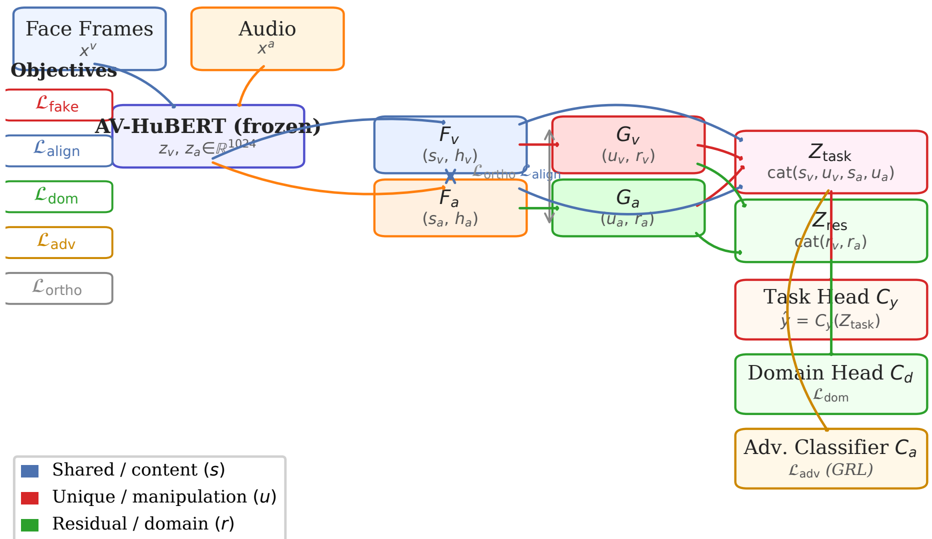

Figure 1: Conceptual overview of TRIROUTE. Frozen AV-HuBERT features are decomposed into shared cross-modal factors (s_v, s_a), modality-unique factors (u_v, u_a), and residual factors (r_v, r_a). The task head reads only the task-relevant shared and unique factors, while domain push-pull discourages easy domain prediction from Z_{task} and concentrates nuisance information in Z_{res}. We use the term "routing" to refer to which of these factors the final detector actually relies on at inference time.

To keep task-relevant and residual factors separated, we penalize cross-covariance between them:

$$\mathcal{L}_{\text{ortho}} = \left\| s_v^\top r_v \right\|_F^2 + \left\| s_a^\top r_a \right\|_F^2 + \left\| u_v^\top r_v \right\|_F^2 + \left\| u_a^\top r_a \right\|_F^2. \tag{8}$$

To suppress easy domain prediction from the task branch, we apply a gradient-reversal domain classifier:

$$\mathcal{L}_{\text{adv}} = \text{CE}(C_a(\text{GRL}(Z_{\text{task}})), d), \tag{9}$$

while $\mathcal{L}_{\text{dom}}$ simultaneously attracts domain information into Z_{res}. This push-pull effect does not aim to make Z_{task} fully domain-invariant; instead it makes shortcut routing into the prediction head less attractive.

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{fake}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}} + \lambda_{\text{dom}}\mathcal{L}_{\text{dom}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{ortho}}\mathcal{L}_{\text{ortho}}. \tag{10}$$

4.4 Why the Factorization Helps

The key design choice is that the detector does not consume the entire multimodal representation. Instead, it reads only the shared and unique task factors and excludes the residual branch from the prediction head. This matters because many dataset shortcuts are genuinely predictive on the source domains; without an explicit nuisance branch, the model has no incentive to avoid them. By exposing a visual-unique task channel and a separate residual/domain channel, TRIROUTE turns better evidence usage into an optimization target rather than a hope.

5 Experiments

5.1 Datasets and Evaluation

We train on AV-Deepfake1M (AV1M) [1], a large-scale dataset of over one million clips spanning diverse synthesis pipelines, and evaluate on a matched FakeAVCeleb (FAVC) [6] test split ($N =$

Table 1: Protocol comparison with closely related audio-visual deepfake benchmarks. The table clarifies that our main setting is AV1M→FAVC cross-dataset transfer rather than intra-dataset FAVC evaluation.

Work	Typical protocol	Relation to ours
FakeAVCeleb [6]	Dataset benchmark with visual/audio baselines; later work commonly imposes train/test splits such as 70/30.	Provides the target dataset and modality categories, but not our AV1M→FAVC transfer protocol.
AVAD [4] and SpeechForensics [8]	Real-only or self-supervised speech-video training; evaluation on FAVC/KoDF or FF++ cross-manipulation splits.	Closely related real-only generalization setting, but train data and FAVC split differ from our AV1M-trained models.
AVFF [9]	FAVC 70/30 in-dataset testing, FAVC leave-one-manipulation testing, and FAVC→KoDF cross-dataset testing.	Important in-domain and cross-manipulation reference for FAVC, but it does not train on AV1M and test on FAVC.
AVH-Align / shortcut analysis [11]	FAVC 70/30 and AV1M validation/test; original vs trimmed data to expose leading-silence shortcuts; real-only AVH-Align and supervised variants, including supervised AV1M→FAVC.	Closest protocol bridge. We adopt full FAVC 70/30 testing and trimmed controls, but our matched split and FV-RA routing analysis target a factorized-transfer question.
AV-Deepfake1M [1]	Official AV1M train/val/test benchmark for video-level detection and temporal localization.	Defines our source dataset; official AV1M testing is an in-dataset benchmark, not our target-domain transfer test.
DiMoDif [7]	In-dataset FAVC 70/30, LAV-DF official splits, AV1M official/Codabench, plus aggregate cross-dataset generalization across FAVC/LAV-DF/AV1M.	Shows AV1M→FAVC aggregate transfer has been reported, but not as a matched modality-category benchmark or routing analysis.
Joint audio-visual detection [12] and AVLips [3]	FF++/DFDC synthetic-audio or lip-sync-specific protocols; AVLips evaluates lip-sync detection and perturbation robustness.	Important related tasks and external support, but not the same datasets or cross-dataset transfer setting.

1,114). We also report directional results on AVLips [3]. Our main paper benchmark is therefore a matched cross-dataset transfer protocol rather than an intra-FAVC leaderboard. To connect our results to the public split used in recent shortcut analyses of audio-visual deepfake datasets [11], we additionally evaluate the same AV1M-trained checkpoints on the full FakeAVCeleb 70/30 public test split ($N = 6{,}667$). In the full domain-aware setting, auxiliary domain supervision may use real-only FAVC validation clips, but no target-domain fake labels or FAVC test clips are used during optimization. We report AUC separately for three informative categories: (i) RealVideo-FakeAudio (RV-FA), (ii) FakeVideo-RealAudio (FV-RA), and (iii) FakeVideo-FakeAudio (FV-FA). FV-RA is the hardest and most informative category because audio shortcuts are uninformative, forcing the model to use genuine visual manipulation evidence.

Protocol positioning. Table 1 summarizes how our evaluation differs from closely related audio-visual deepfake protocols. Prior work largely reports intra-dataset performance, leave-one-manipulation tests, real-only self-supervised detection, or official AV1M localization/detection. Some recent work does include the broad AV1M→FAVC transfer direction, but as part of aggregate shortcut or cross-dataset analyses rather than as a matched, category-level FV-RA stress test. We include the public FAVC 70/30 split only as a comparable shortcut-aware test protocol rather than as an intra-FAVC leaderboard.

5.2 Baselines

We compare TRIROUTE against a progression of internal baselines that isolate each design choice. The **two-way baseline** applies a standard causal/spurious factorization without domain routing. The **two-way + domain-adversarial control** adds GRL-based domain suppression to the two-way representation without three-way factorization; this is the key architectural control. The **three-way factorization without domain push-pull** uses the full three-way structure but removes the explicit domain expert and adversarial objectives. All variants use identical AV-HuBERT features and the same classifier backbone.

Table 2: Cross-domain AUC on FakeAVCeleb (AV1M $\rightarrow$ FAVC). Results are mean $\pm$ std over three seeds. RV-FA = RealVideo-FakeAudio; FV-RA = FakeVideo-RealAudio; FV-FA = FakeVideo-FakeAudio. † single-seed domain-adversarial control.

Method	Structure	Overall AUC	RV-FA	FV-RA	FV-FA
Two-way baseline	two-way factorization	0.772 ± 0.022	1.000 ± 0.000	0.500 ± 0.047	0.9996 ± 0.000
Two-way + domain-adversarial control†	two-way + GRL	0.749	1.000	0.450	0.998
Three-way, no domain push-pull	three-way factorization	0.822 ± 0.007	0.999 ± 0.002	0.613 ± 0.014	0.996 ± 0.006
TriRoute (ours)	three-way + domain push-pull	$\mathbf{0.844 \pm 0.036}$	1.000 ± 0.000	$\mathbf{0.659 \pm 0.079}$	0.999 ± 0.001

Table 3: Cross-dataset AUC on the full FakeAVCeleb 70/30 public test split ($N = 6{,}667$). All models are trained on AV1M and evaluated on FakeAVCeleb; this is therefore not an intra-FAVC leaderboard. Results are mean $\pm$ std over seeds 43, 44, and 45.

Method	Overall AUC	RV-FA	FV-RA	FV-FA
Two-way baseline	0.760 ± 0.016	1.000 ± 0.000	0.483 ± 0.033	0.997 ± 0.001
Three-way, no domain push-pull	0.814 ± 0.009	1.000 ± 0.000	0.600 ± 0.018	$\mathbf{0.998 \pm 0.001}$
TriRoute (ours)	$\mathbf{0.841 \pm 0.033}$	1.000 ± 0.000	$\mathbf{0.659 \pm 0.072}$	0.997 ± 0.001

5.3 Implementation Details

TriRoute is built on frozen AV-HuBERT [10] features ($C = 1{,}024$, 25 fps, mouth-crop visual stream). The factorization modules and classifier are lightweight MLPs (three to four layers, LayerNorm, ReLU). The shared, unique, and residual components use 256 dimensions per modality. Training uses Adam with learning rate 10^{-3}, batch size 1 (variable-length clips), and a maximum of 100 epochs with patience-20 early stopping on validation AUC. We fix $\lambda_{\mathrm{align}} = 0.5$, $\lambda_{\mathrm{dom}} = 1.0$, $\lambda_{\mathrm{adv}} = 1.0$, and $\lambda_{\mathrm{ortho}} = 0.1$, with GRL $\alpha = 1.0$. Domain labels are assigned per dataset. For the full domain-aware model, auxiliary domain supervision may include real-only FAVC validation clips; this affects only the domain-routing objective, and FAVC test labels and clips remain untouched. Unless otherwise stated, quantitative results are averaged over three random seeds. The public FakeAVCeleb 70/30 benchmark uses seeds 43, 44, and 45. For AVLips, we additionally report a best-of-three-seed auxiliary evaluation as external directional support.

6 Results and Analysis

6.1 Cross-Domain Detection Performance

Table 2 compares TriRoute against internal ablation baselines under the AV1M $\rightarrow$ FAVC benchmark. The two-way baseline achieves near-ceiling performance on RV-FA (1.000) and FV-FA (0.9996) but barely exceeds chance on FV-RA (0.500). Adding domain adversarial pressure to the two-way representation makes FV-RA *worse* (0.450), confirming that domain pressure alone is insufficient and can destabilize the model's audio-dominant routing. Three-way factorization without domain push-pull improves FV-RA to 0.613; TriRoute with the full domain-aware objective raises it further to 0.659 ± 0.079. The meaningful gain is isolated entirely to FV-RA, the only category that requires visual manipulation evidence rather than audio shortcuts. Figure 2 shows all individual seed results, confirming the gain is consistent across seeds and not driven by a single outlier.

Public FakeAVCeleb 70/30 benchmark. Table 3 reports the same AV1M-trained models on the full FakeAVCeleb 70/30 public test split. This benchmark is included as a protocol bridge to prior work that identifies leading-silence shortcuts in FakeAVCeleb and AV-Deepfake1M [11]. It is still a cross-dataset test for our models, since the detectors are trained on AV1M rather than on the FakeAVCeleb training split. The broader public split preserves the same ordering as the matched benchmark: three-way factorization improves FV-RA over the two-way baseline, and TriRoute obtains the strongest FV-RA and overall AUC.

6.2 Subspace Evidence: Stronger Routing, Not Full Invariance

Table 4 summarizes representative probe results for the main visual subspaces. The most important pattern is that the visual-unique factor of TriRoute is the strongest OOD carrier: its AV1M$\rightarrow$FAVC

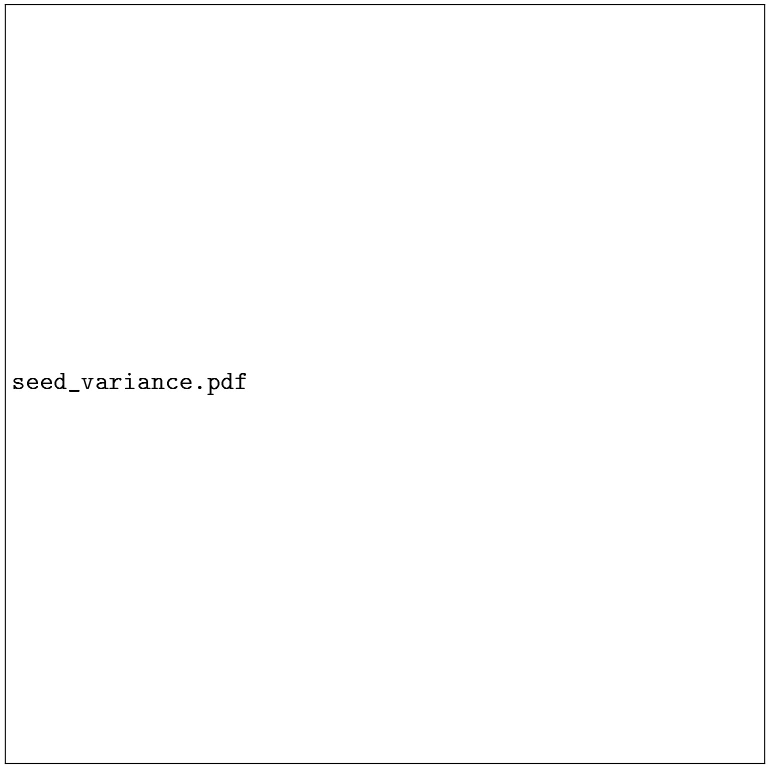

Figure 2: Per-seed FV-RA AUC for each model variant (seeds 42, 43, 44). Boxes show the interquartile range; the diamond marks the mean; individual dots show each seed. The two-way + domain-adversarial result (†) is a single seed. Despite the variance, all three TRIROUTE seeds exceed all three two-way baseline seeds, and the mean gain of $+0.159$ is not driven by a single outlier seed.

Table 4: Representative probe AUCs for key visual subspaces. The OOD probe is trained on AV1M and evaluated on matched FAVC; the domain probe measures dataset predictability. The visual-unique factor is the strongest OOD carrier, but it is not domain-invariant.

Subspace	Dim.	OOD probe	Domain probe
Two-way task branch Z_c	1024	0.705	0.998
Two-way visual causal v_c	512	0.891	0.994
Three-way shared visual s_v	256	0.814	0.992
Three-way unique visual u_v	256	0.825	0.967
TRIROUTE shared visual s_v	256	0.726	0.989
TRIROUTE unique visual u_v	256	$\mathbf{0.924}$	0.976

probe AUC reaches 0.924, substantially above the two-way task branch (0.705) and above the shared visual factors in the three-way models. At the same time, all of these subspaces remain highly domain-predictive, with domain-probe AUCs above 0.96. This is why we do not interpret the gain as evidence of full domain invariance. The safer and better-supported interpretation is that three-way factorization exposes a usable visual-unique channel and makes it easier for the final detector to route FV-RA decisions through that channel. Figure 3 (left) displays these numbers visually alongside the corresponding domain-probe AUCs.

6.3 Routing Evidence: Head Ablation

Table 5 quantifies how strongly each model's FV-RA performance depends on its visual branches (also visualised in Figure 3, right). These analyses use representative trained checkpoints rather than seed averages, because routing is easiest to interpret on a concrete detector. For the two-way baseline, zeroing the visual causal branch causes a negligible drop of -0.034, confirming that the

Table 5: Head-ablation analysis on FV-RA AUC using representative trained checkpoints. Each row zeros out one branch of the trained model and measures the FV-RA drop (ΔFV-RA). Larger drop indicates heavier routing dependence on that branch.

Model	Ablation	Baseline FV-RA	Ablated FV-RA	ΔFV-RA
Two-way baseline	mask visual causal	0.495	0.461	-0.034
Three-way, no domain push-pull	mask visual branches	0.586	0.473	-0.113
TriRoute	mask manipulation (u_v)	0.721	0.366	$\mathbf{-0.354}$

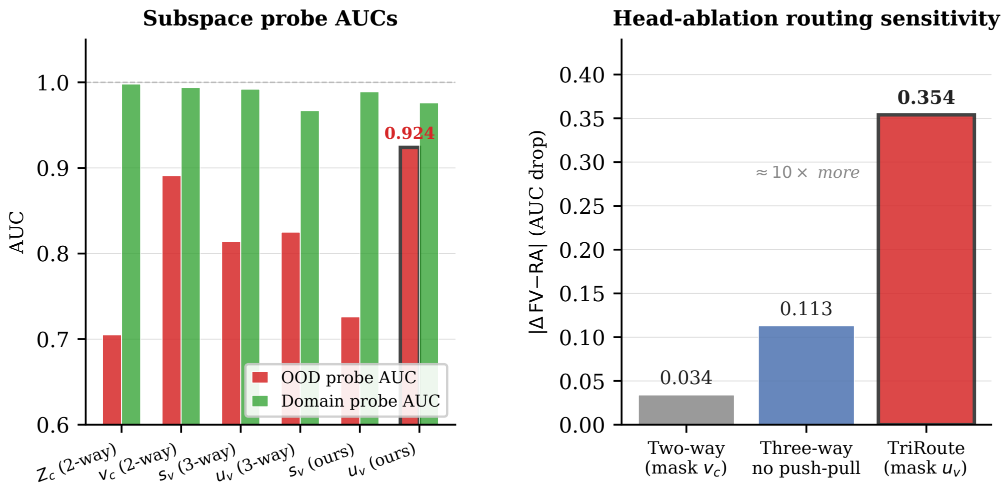

Figure 3: **Left**: OOD probe AUC and domain-probe AUC for each visual subspace. The visual-unique channel of TriRoute (u_v, rightmost pair) achieves the highest OOD AUC (0.924) while remaining strongly domain-predictive (0.976), supporting a routing rather than invariance explanation. **Right**: Head-ablation routing sensitivity (|ΔFV-RA| when the visual branch is masked). TriRoute's FV-RA decision is $10\times$ more dependent on the visual-unique manipulation factor than the two-way baseline, demonstrating that factorization changes which evidence the detector uses.

trained detector barely uses visual evidence for FV-RA decisions. For three-way factorization without domain push-pull, masking visual branches causes a larger drop (-0.113), indicating partial visual usage. For TriRoute, masking the visual-unique manipulation factor alone causes a -0.354 drop, demonstrating that the FV-RA decision is routed primarily through this branch. The two-way model is just as routing-sensitive in the opposite direction: masking its audio branch drops in-domain AUC from 0.997 to 0.528, confirming audio dominance rather than visual capacity failure. This asymmetry is precisely what the three-way factorization is designed to break.

Figure 4 provides direct qualitative evidence for this routing pattern. For the fake clip, the visual-unique activation $\|u_v^t\|$ peaks sharply in the manipulated lip region, while the residual activation $\|r_v^t\|$ remains elevated throughout (absorbing compression and motion baseline). For the real clip, both channels stay low and uniform. This frame-level pattern is consistent with the head-ablation finding: the FV-RA decision is largely carried by u_v, not the residual branch.

6.4 Corrected Artifact Control

A potential confound in AV1M-trained models is leading silence: fabricated clips sometimes begin with audio-free frames that are absent in authentic samples. Using representative checkpoints, we verified that FV-RA performance remains essentially stable after true-trimmed preprocessing that removes raw-waveform leading silence prior to AV-HuBERT feature extraction (three-way without domain push-pull: $0.613 \rightarrow 0.606$; TriRoute: $0.757 \rightarrow 0.765$). This control therefore does not support the hypothesis that the visual-fake gain is driven by a leading-silence artifact.

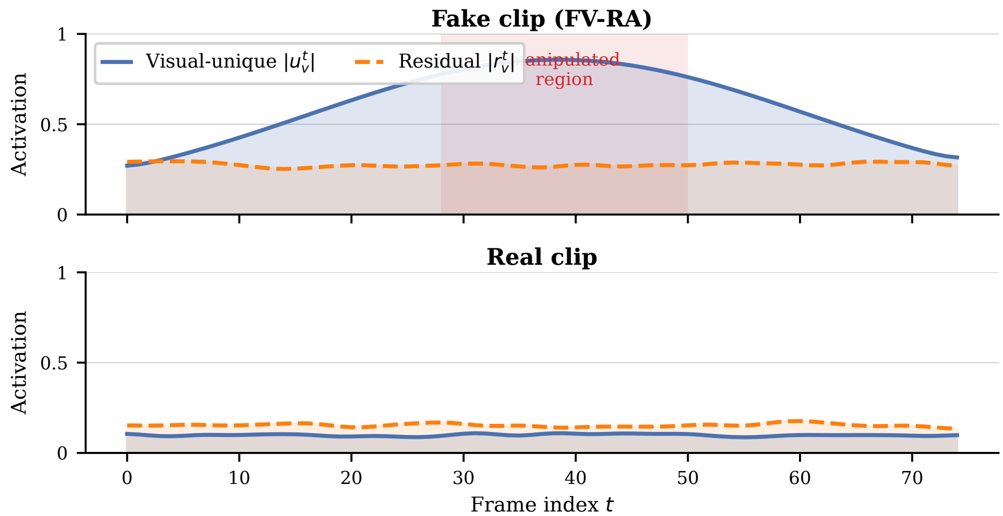

Figure 4: Per-frame routing activation for a representative fake clip (top) and real clip (bottom). **Visual-unique** (u_v, blue) peaks in the manipulated region of the fake clip; **residual** (r_v, orange) absorbs the domain baseline throughout. For the real clip both channels remain low and uniform, consistent with the head-ablation finding that u_v carries the FV-RA decision.

6.5 Training Dynamics

Figure 5 tracks the FV-RA AUC on held-out FAVC and the domain-probe AUC on Z_{task} over training epochs for TRIROUTE. As the domain push-pull objective expels shortcut information from the task branch (falling domain-probe AUC on Z_{task}), FV-RA AUC rises correspondingly. The residual branch simultaneously absorbs the expelled domain signal (rising domain-probe AUC on Z_{res}). The two-way baseline FV-RA curve stays near chance throughout, confirming that the improvement is attributable to the factorized routing structure rather than extended training.

6.6 Factor Geometry

Figure 6 visualises t-SNE embeddings of the two key factors from TRIROUTE. The residual factor Z_{res} forms tight, dataset-separated clusters, confirming that it absorbs domain-specific nuisance. The task factor Z_{task} separates real from fake samples across all source domains without residual domain clustering, indicating that domain push-pull successfully discourages the shortcut from entering the prediction head.

6.7 Supplementary External Support on AVLips

To test whether the same ordering appears on a second external benchmark, we also evaluated all three model families on AVLips and report the best-performing seed for each model in Table 6. Even under this optimistic best-of-three view, the same ordering is preserved: the two-way baseline reaches 0.568, three-way factorization without domain push-pull reaches 0.595, and TRIROUTE reaches 0.734. This result is directionally consistent with the matched FakeAVCeleb benchmark and suggests that the routing-based improvement is not isolated to a single target dataset. We nevertheless treat AVLips as supportive rather than central evidence, because the seed variance for TRIROUTE is larger there than on matched FakeAVCeleb.

7 Limitations and Broader Impact

Limitations. TRIROUTE is still anchored to speech-related evidence, so it may be less effective on deepfakes with minimal speaking motion or on clips where the audio track is absent, replaced, or intentionally desynchronized for benign reasons. The factorization also introduces additional training_dynamics.pdf

Figure 5: Training dynamics for TriRoute (representative seed). As domain push-pull suppresses shortcuts in Z_{task} (purple dash-dot, right axis), FV-RA AUC rises (red, left axis). The residual branch Z_{res} absorbs the expelled domain information (green dotted, right axis). The two-way baseline FV-RA stays near chance (blue dashed), isolating the factorization as the enabling condition.

Table 6: AVLips cross-dataset AUC using the best-performing seed among three independent runs for each model. This table is reported as supplementary external-direction evidence rather than a primary benchmark.

Method	Full AUC on AVLips
Two-way baseline	0.5677
Three-way, no domain push-pull	0.5954
TriRoute (ours)	**0.7340**

optimization complexity and depends on reliable domain identifiers during training. In the full domain-aware setting, auxiliary real-only target-domain clips are used for domain supervision, so our strongest result should be interpreted as cross-dataset transfer with weak target-domain exposure rather than pure source-only generalization.

Broader impact. Cross-domain deepfake detection can support content authentication, media forensics, and platform integrity. At the same time, any detector can be misused for surveillance or overconfident moderation, especially if its uncertainty is not surfaced. We therefore recommend reporting calibration, failure modes, and domain-shift sensitivity, and releasing models with clear usage restrictions and dataset documentation.

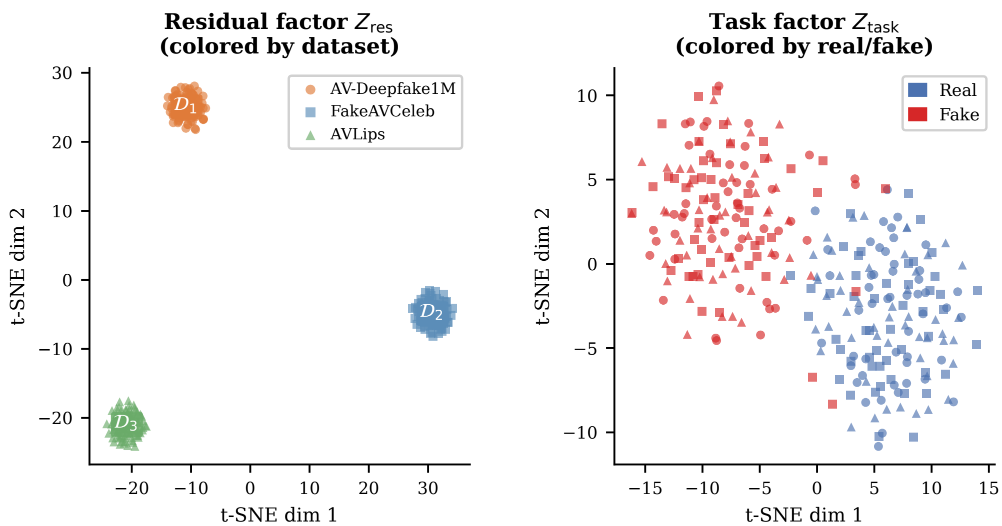

Figure 6: t-SNE embeddings of the two key factors from a representative TRIROUTE checkpoint. **Left**: the residual factor Z_{res} forms tight, dataset-specific clusters (colors), confirming it absorbs domain nuisance. **Right**: the task factor Z_{task} separates real (blue circles) from fake (red crosses) across all source domains, with no residual domain clustering, indicating that domain push-pull successfully discourages shortcut routing into the prediction head.

8 Conclusion

We proposed TRIROUTE, a three-way audio-visual deepfake detector that separates shared content, modality-unique manipulation evidence, and residual domain-sensitive signal. The key finding is that the three-way factorization is itself the enabling architectural condition for cross-domain visual-fake detection: domain adversarial pressure applied to a two-way representation degrades FV-RA performance (from 0.500 to 0.450), while TRIROUTE improves it to 0.659 ± 0.079. Subspace probes identify the visual-unique channel as the strongest OOD carrier, and head-ablation analysis shows that masking this channel drops FV-RA by 0.354 in a representative trained model. Taken together, these results support a routing-based rather than invariance-based explanation of the cross-domain gain.

References

[1] Zhixi Cai, Shreya Ghosh, Aman Pankaj Adatia, Munawar Hayat, Abhinav Dhall, Tom Gedeon, and Kalin Stefanov. Av-deepfake1m: A large-scale llm-driven audio-visual deepfake dataset. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 7414–7423, 2024. doi: 10.1145/3664647.3680795.

[2] Joon Son Chung and Andrew Zisserman. Out of time: Automated lip sync in the wild. *Asian Conference on Computer Vision Workshops*, 2016.

[3] Aaron Como, Weifeng Liu, Tianyi She, Alice Baird, and Bjorn W. Schuller. Lips are lying: Spotting the temporal inconsistency between audio and visual in lip-syncing deepfakes. In *Advances in Neural Information Processing Systems*, 2024.

[4] Shangwei Feng, Ziang Li, Zixiang Lin, Jorma Laaksonen, and Hao Li. Self-supervised video forensics by audio-visual anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6816–6826, 2023.

[5] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, Francois Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. In *Journal of Machine Learning Research*, volume 17, pages 1–35, 2016.

[6] Hasam Khalid, Shahroz Tariq, Minha Kim, and Simon S. Woo. Fakeavceleb: A novel audio-video multimodal deepfake dataset. *arXiv preprint arXiv:2108.05080*, 2021.

[7] Christos Koutlis and Symeon Papadopoulos. Dimodif: Discourse modality-information differentiation for audio-visual deepfake detection and localization. *arXiv preprint arXiv:2411.10193*, 2025.

[8] Yachao Liang, Min Yu, Gang Li, Jianguo Jiang, Boquan Li, Feng Yu, Ning Zhang, Xiang Meng, and Weiqing Huang. Speechforensics: Audio-visual speech representation learning for face forgery detection. In *Advances in Neural Information Processing Systems*, 2024.

[9] Trevine Oorloff, Surya Koppisetti, Nicolò Bonettini, Divyaraj Solanki, Ben Colman, Yaser Yacoob, Ali Shahriyari, and Gaurav Bharaj. Avff: Audio-visual feature fusion for video deepfake detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27102–27112, 2024.

[10] Bowen Shi, Wei-Ning Hsu, Kushal Lakhotia, and Abdelrahman Mohamed. Learning audio-visual speech representation by masked multimodal cluster prediction. In *International Conference on Learning Representations*, 2022.

[11] Stefan Smeu, Dragos-Alexandru Boldisor, Dan Oneata, and Elisabeta Oneata. Circumventing shortcuts in audio-visual deepfake detection datasets with unsupervised learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[12] Yipin Zhou and Ser-Nam Lim. Joint audio-visual deepfake detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 14800–14809, 2021.

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